

Composite Timing Metrics Do Not Measure Warning Capability: A Video-Blind Constant Matches the Eighth of Thirteen Teams, and the Corrected Protocol Needs Correcting

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Abstract

Traffic-accident anticipation is increasingly scored by composite metrics that add an earliness term — time-to-accident at a fixed risk threshold — to discrimination terms such as average precision and area under the ROC curve. The composition fails structurally: the discrimination terms are bounded on the unit interval, the earliness terms scale with clip length, and the earliness terms are maximised by a constant risk score above the threshold — a submission that, deployed, would warn continuously. We measure the consequence twice. On a live benchmark of 1,417 dashcam clips and thirteen teams, whose labels are withheld and whose metric weights are undisclosed, a constant risk score of 0.51 that reads no pixels scores 2.35291: it matches the eighth-placed team at the precision the leaderboard displays and exceeds five of thirteen entries including our own. Twenty-five such submissions, designed so the undisclosed terms cancel under differencing, decompose the metric without any label: discrimination contributes 0.35105 and the two timing terms 2.00186, 5.70 times as much. The same probes establish that four curves crossing the threshold at the same frame score 0.091 apart, a spread the published definition cannot produce, so the stated formula does not reproduce its own scorer. We then validate a corrected protocol on an independent corpus of 1,500 annotated clips, half containing no collision. The pathology reproduces — a constant takes 99.9% of the available earliness at exactly chance discrimination — and the cost it hides becomes measurable: such a warning alarms for 60 minutes in every hour of ordinary driving. The protocol behaves as predicted but is then beaten by a control that reads only the video file’s header, because clip-level conditioning cannot see a curve that is flat within a clip. We replace it with a frame-level statistic, coverage of the annotated actionable window against alarm duty cycle, under which the video-blind family scores zero and a sighted model does not. Our own ninth-placed entry, which scores below the constant, is the case study.

Index Terms— Accident anticipation, evaluation methodology, benchmark design, video-blind baseline, time-to-accident, collision warning systems.

1 Introduction

Road traffic injury kills approximately 1.19 million people a year by the World Health Organization’s count [1]. Many of those deaths are foreseeable in the seconds before they occur, which is why systems that anticipate a collision — rather than detect one after the fact — are pursued as a countermeasure, and why a growing literature reports how early such systems can raise an alarm [2]. Whether a reported anticipation result establishes that a system could actually warn a driver is a question about measurement, and it is the question this paper addresses.

Anticipation is increasingly scored by a composite metric: a weighted sum of discrimination terms — average precision (AP) and area under the ROC curve (AUC) — and an earliness term, the interval between alarm and collision read at a fixed risk threshold. The composition is intuitive. An anticipator should separate accident clips from ordinary driving and should raise the alarm early, and summing terms that measure each seems a reasonable way to ask for both.

It is not, and the reason is a difficulty of type rather than of tuning. AP and AUC are bounded on the unit interval; an earliness term counted in frames is bounded only by the length of the clip. Summing a bounded quantity with an unbounded one lets the unbounded one decide the ranking, and makes that ranking depend on how the corpus was windowed. Worse, an earliness term read at a threshold has a trivial global maximiser: a risk score that exceeds the threshold at frame 0 and never falls attains, on every clip, the largest value that clip admits. No method that reads the video can exceed it. A method can only match it, and then compete on the bounded half.

That argument follows from the definitions alone. What has been missing is a measurement of what it costs in practice, and the measurement is awkward to obtain: the benchmarks using these metrics withhold their labels, and the competitions using them do not publish the weights.

This paper supplies it twice over. On a live accident-anticipation benchmark — 1,417 dashcam clips, thirteen teams, a scorer we never saw — a constant risk score of 0.51 that reads no pixels scores 2.35291 against a winning

2.50165, matching the eighth-placed team at the precision the leaderboard displays and exceeding five of thirteen entries. A family of submissions that likewise read nothing, designed so that the undisclosed terms cancel when two are differenced, then recovers the metric’s parts from outside the competition: the timing terms are worth **5.70 times** what discrimination contributes to such a submission, and the score falls at almost exactly one sixtieth of a point for every frame of delay.

Because that benchmark publishes no labels, it can show the metric being exploited but cannot show anything being fixed. We therefore repeat the measurement on an independent, fully annotated corpus of 1,500 clips, half of which contain no collision. There the pathology reappears — a constant takes 99.9% of the available earliness at exactly chance discrimination — and, for the first time, the cost of the behaviour it rewards becomes measurable: a warning that maximises this metric alarms for **60 minutes in every hour** of ordinary driving.

The safety consequence is therefore not that the metric is untidy. The submission maximising its earliness half is a warning that never turns off. Whether a driver comes to trust a collision warning or to ignore it depends on when it arrives relative to what the driver expects [3], and a warning that never stops carries no timing information at all. The metric awards that behaviour its maximum and cannot do otherwise, because its timing terms are defined only on clips where an accident occurs, so the burden such a system imposes on ordinary driving falls entirely on the discrimination half.

We came to the question the awkward way: we built a training-free ensemble, entered it in that competition, and placed ninth of thirteen. The paper is not about that system — it scores below the constant — but it is what made the metric worth examining, and it appears here as the case study.

Contributions. We report a measured demonstration on a live benchmark: a video-blind constant matches the eighth-placed private score and exceeds five of thirteen teams, our own included. We give a black-box decomposition, in which video-blind probes built so that unknown terms cancel under differencing separate a metric whose weights are undisclosed and whose labels are withheld, recovering the discrimination floor, both timing terms, the slope in the crossing frame and the accident-onset distribution. We show that the published definition does not reproduce the scorer: four curves crossing at the same frame, none dipping below it afterwards, score 0.091 apart, where the stated formula requires equality. We then validate the corrected protocol where ground truth exists; its predicted behaviour holds, but it is beaten by a control reading only the video file’s header, so we replace its clip-level conditioning with a frame-level alarm statistic and show that this orders submissions by what they can see. Every leaderboard score was computed by the competi-

tion’s scoring service against ground truth we do not have; derived quantities are our arithmetic on those scores and are marked as such.

One point of scope, because this paper prints a number beside other teams’ numbers. We make no claim about any other submission or published method: we did not re-run anyone’s system, and could not have evaluated it if we had, because the labels are withheld from every entrant. Where a leaderboard score is numerically identical to a control’s, we state that arithmetic fact and draw one inference — the metric assigns the two submissions the same value. A whole family of curves receives it, which is precisely the property under examination.

2 Background

2.1 Where the metrics come from

Vision-based accident anticipation was posed in its modern form by Chan *et al.* [4], who introduced the Dashcam Accident Dataset together with the measure the field still reports: the interval between the first frame at which a risk score crosses a threshold and the annotated onset of the collision. Bao *et al.* [5] contributed the Car Crash Dataset and an uncertainty-based formulation, and Karim *et al.* [6] gave the reporting convention most subsequent work adopts: average precision, mean time-to-accident over thresholds, and time-to-accident at 80% recall.

It is worth stating plainly what that convention gets right, because this paper is about what happens when it is set aside. It reports a discrimination metric alongside every timing metric, and it reads timing *at a fixed operating point* of the discrimination metric. A method therefore cannot buy time-to-accident by alarming indiscriminately: the recall constraint holds it to a fixed false-alarm behaviour before its timing is counted at all. The repair proposed in Section 7 is in large part a return to that discipline.

Architectures built on this foundation run from supervised convolutional and temporal models [4, 5, 6] through self-supervised formulations that reduce the dependence on scarce collision labels [7, 8, 9], to training-free pipelines pairing a vision-language model [10] with a motion signal [11, 12, 13] — though those target detection or classification and none reports a timing metric.

Two lines of work deserve particular attention, because they anticipate part of this paper’s subject. Pjetri *et al.* [14] train with an objective requiring the predicted danger score to be non-decreasing as the collision approaches, and Zou *et al.* [15] introduce an adaptive monotonic constraint loss with the same intent. Both are well-motivated responses to score oscillation and we criticise neither. But both illustrate the hazard examined here: a risk score constrained not to fall is, by construction, closer to the maximiser of a threshold-crossing timing metric than a score

free to fall. The metric cannot separate the benefit of the constraint from the artefact of it, and neither paper reports a control that would.

2.2 From a convention to a single number

Competitions need something the convention of [6] does not supply: a total order over submissions. The natural way to obtain one is to sum the metrics, and that is what the benchmark studied here does [16], adding two threshold-crossing timing terms to average precision and area under the ROC curve.

Summing is where the difficulty enters, and it is a difficulty of type rather than of tuning. Average precision and area under the curve are bounded on the unit interval; a time-to-accident measured in frames is bounded only by the length of the clip. Adding them makes the sum depend on how the corpus was windowed and lets the unbounded term decide the ranking. Section 6 develops the argument; Sections 4 and 5 measure its magnitude.

2.3 The missing control

The check that exposes such a problem is a control that does not read the input, and its value has been established repeatedly elsewhere: dataset signatures strong enough to identify a photograph’s source [17]; models given only half the input scoring far above chance on standard inference corpora [18, 19], after which partial-input baselines became standard practice; and tuned simple baselines matching most recently published neural recommenders [20]. In each case the control was cheap, and in each case nobody had run it.

Accident anticipation has every precondition those episodes had. It reports a timing metric whose maximiser is trivial, it is moving toward composite single-number scores, and its newest benchmarks withhold labels — so an entrant cannot compute the discrimination half at all and must develop against a locally computable proxy, the only candidate being the shape of one’s own risk curve, which a constant maximises.

The pathology has not gone unnoticed, but it has been noticed as an anomaly rather than isolated. Zhao *et al.* [21] observe that the standard calculation credits a false alarm in a safe scene with the interval to some later, causally unrelated collision. Goldshmidt *et al.* [22] report baselines making physically implausible predictions nine to ten seconds ahead. Most directly, Caselli *et al.* [23] publish a table in which two established methods record areas under the curve of 45.5 and 41.0 — below chance — beside mean times-to-accident of 10.216 and 10.263 seconds. A method that cannot separate accident clips from ordinary ones is there credited with ten seconds of anticipation: this paper’s argument, already legible in someone else’s results table. None of these accounts isolates the effect, and none reports what a submission that reads nothing would score.

The nearest methodological precedent lies in an adjacent task. Korkut *et al.* [24] audit four traffic video question-answering benchmarks by running vision–language models blind, and find that removing the video consistently improves accuracy on one of them. That is a control of exactly the kind advocated here, applied to multiple-choice accuracy rather than to per-frame risk. We are not aware of an equivalent control reported for the timing and discrimination metrics used in anticipation, which is the gap this paper fills.

We state the scope of that observation precisely: it covers the works cited here whose results tables we inspected directly. Four we consulted are behind paywalls we could not open, and we exclude them rather than assume. It is not a systematic audit of the field and we do not present it as one.

2.4 What is and is not new here

Monotonicity constraints on risk scores are prior work [14, 15] and this paper claims none; so are training-free pipelines pairing a vision–language model with a motion signal [11]. Nor is the general idea that a benchmark can be satisfied by a degenerate submission new, and the suspicion that time-to-accident is inflatable has been voiced before [21, 22, 23], as has a blind control on an adjacent traffic-video task [24]. What is new is the measurement of how far it goes on this class of metric, a probe methodology that decomposes such a score from outside the competition holding the labels, and an evaluation protocol that has been tested against adversarial controls, found insufficient, and repaired.

3 The Benchmark, the Controls and the Probes

This section describes the benchmark whose metric is under examination and the two families of submission with which we measured it. Neither family reads a video frame.

3.1 The competition

Measurements were made on Zero-shot Accident Anticipation, a public competition hosted on Kaggle [16]. It ran from 17 February to 15 April 2026, drawing 59 entrants of whom 20 submitted, forming 13 teams across 194 submissions. Late submission remains open and is scored against the same splits, which is how the experiments of Section 4 were run after the closing date. Every figure quoted here was read from the competition’s own pages on 31 August 2026.

Two properties make it a suitable instrument. The organisers hold the labels and compute every score, so nothing reported here depends on our own implementation of a metric. And a submission is a plain list of numbers per

clip, so it need not contain a model at all — which is what makes a video-blind control submittable.

The corpus is a curated 1,417-clip subset of MM-AU [25], every clip exactly 150 frames at 30 fps. The published test file carries an identifier, a video identifier, a start and end frame, and a caption. There is no label, no accident time and no alert time.

One consequence deserves emphasis, because the rest of this paper turns on it. Average precision, area under the ROC curve, a false-positive rate, and any time-to-accident measured to a true onset are *not computable by any entrant*. An entrant wanting to measure progress locally must substitute something, and the only quantities computable from a submission alone are properties of the risk curve’s own shape — the frame at which it crosses the threshold, whether it falls back, how steeply it rises. We did exactly this, and Section 4 measures what that proxy was worth. A second consequence we identify without quantifying: the 79 distinct captions describe outcomes (“lead vehicle stops”, 146 clips; “a vehicle controls loss”, 144), and they are supplied at inference time, so a method conditioned on them has access to the outcome it is asked to anticipate.

3.2 The metric, as published

The evaluation page defines the score as a weighted sum of four quantities,

$$\text{score} = w_{\text{AP}}\text{AP} + w_{\text{AUC}}\text{AUC} + w_{\text{TTA}}\text{TTA}@0.5 + w_{\text{STTA}}\text{STTA}@0.5 \quad (1)$$

with the weights fixed by the organisers and not disclosed. Positives are accident clips, negatives ordinary driving. The timing terms are read at a risk threshold of 0.5:

$$\text{TTA}@0.5 = \max\{t_{ai} - t_a \mid p_{t_a} > 0.5, 0 \leq t_a \leq t_{ai}\} \quad (2)$$

$$\text{STTA}@0.5 = \max\{t_{ai} - t'_a \mid p_t > 0.5 \forall t \in [t'_a, t_{ai}]\} \quad (3)$$

where t_{ai} is the accident start frame and t_a the first frame at which the risk score exceeds the threshold. STTA additionally requires the score to remain above the threshold continuously from t'_a to t_{ai} .

Three observations follow from these definitions alone, before any measurement.

First, AP and AUC are bounded on the unit interval, while TTA and STTA are counted in frames and bounded only by t_{ai} — hence by how the corpus was windowed. The four terms are not commensurable, and their sum inherits the scale of the larger pair.

Second, both timing terms are maximised by the same trivial submission. A risk score exceeding 0.5 at frame 0 and never falling has $t_a = 0$ and satisfies the continuity condition on every clip, attaining $\text{TTA} = \text{STTA} = t_{ai}$, the largest value each clip admits. No submission can exceed it; one can only match it, and then compete on AP and AUC.

Third, neither timing term reads the magnitude of the score, only whether it stands above 0.5. A submission that is barely committed and one that is certain receive identical timing credit, so the metric offers no incentive to calibrate.

3.3 Video-blind controls

A video-blind submission is one whose risk curve is a function of the frame index alone: identical for every clip, computed without opening a frame. We use a family rather than a single curve, because one flattering curve invites the objection that the comparison was chosen to flatter it. The family comprises constants at 0.51 and 0.99; a curve held at 0.49, which never crosses; a linear ramp; logistic curves centred at frames 60, 75, 90 and 105; step functions rising from 0.49 to 0.51 at frames 0, 10, 25, 50, 75, 100, 125 and 140, plus a calibration step at frame 76; a curve that crosses at frame 0 then drops back, and one that crosses and dips across frames 50–59; and two graded curves. That is 21 distinct shapes submitted as 25 files — four were sent twice, and the repeats agree to within 3×10^{-5} , which is a check on the scoring service and not only on our bookkeeping.

Before spending a submission slot we regenerated the organisers’ sample file from its own parsed values and compared it byte for byte against the distributed file: identical. A scoring anomaly therefore cannot be attributed to our file writer.

3.4 Probes that isolate the terms

The decomposition rests on one observation. Every video-blind submission is identical across clips, so — provided the scorer reduces each curve to one clip-level score by any deterministic function of that curve — every member of the family produces an all-way tie among clips, and therefore earns identical AP and identical AUC. Whatever those terms are worth, they are worth the same to all of them, and *they cancel exactly in any difference between two members*.

That turns the leaderboard into an instrument. Three probes suffice. A curve held at 0.49 never crosses, so both timing terms are zero and its score is the discrimination floor alone. A curve at 0.51 in frame 0 and 0.49 thereafter crosses once and then falls, earning the time-to-accident term but forfeiting the stable one. A constant 0.51 earns both. Differencing the three gives each term separately. A fourth probe, holding 0.51 throughout except for a dip across frames 50–59, moves the stable term’s reference point while leaving the first crossing untouched, and serves as a consistency check.

The step family measures the score’s dependence on the crossing frame directly, giving a second and independent route to the same quantity. Two routes to one number is what makes the result safe to report.

Table 1: The final private leaderboard as published, with this work’s video-blind controls below the rule. The controls were submitted after the closing date and scored by the same service against the same private split; they are not leaderboard entries and are not ranked.

rank	entry	private score
1	CVLAB	2.50165
2	BUPT MIC Lab	2.46234
3	Tianhao Zhao	2.42361
4	cr-tfx	2.41555
5	wtiaw_tiaw	2.39725
6	IsaacFI	2.39027
7	yyttl	2.36844
8	Paulini38	2.35291
9	<i>our entry</i>	2.00585
10	Song_Ren	1.46008
11	WDL	1.35992
12	YooHyun.2	1.04203
13	Ratnachand Kancharla	1.03913
—	benchmark row on the leaderboard	0.58333
<i>video-blind controls (this work)</i>		
—	constant 0.51 at every frame	2.35291
—	constant 0.99 at every frame	2.35291
—	linear ramp 0.000 $\rightarrow$ 1.000	1.06725
—	organisers’ sample, resubmitted	1.04203
—	never crosses (0.49 throughout)	0.35105

4 Results on the Live Benchmark

Every score in this section was computed by the competition’s scoring service against ground truth we do not hold. We implemented none of the four metrics. Quantities derived from those scores — the decomposition, the fit, the ratios — are our own arithmetic and are identified as such.

4.1 A video-blind constant on the official leaderboard

Table 1 reproduces the final private leaderboard as published, with the video-blind controls below the rule: submitted after the closing date and scored by the same service against the same private split. They are not leaderboard entries.

A constant risk score of 0.51 scores **2.35291**. That equals the eighth-placed team’s score at the five decimals the leaderboard displays, exceeds five of the thirteen teams including our own, and falls 0.14874 short of the winner. Per the scope stated in Section 1, the only inference we draw from the coincidence at eighth place is that the metric assigns those two submissions the same value; a whole family of curves — every curve above 0.5 at frame 0 that never dips — receives it, and we infer nothing about what any team submitted.

Table 2: Probes designed so the discrimination terms cancel under differencing, and the components recovered from them. The ratio compares the timing terms at their maximum against discrimination at chance.

probe	private score	
never crosses (0.49 throughout)	0.35105	
crosses at frame 0, then drops	1.22698	
crosses at frame 0, dips 50–59	1.85347	
constant 0.51	2.35291	
component	contribution	
AP + AUC, to a video-blind submission	0.35105	(1.00 $\times$)
TTA@0.5, saturated	0.87593	(2.50 $\times$)
STTA@0.5, saturated	1.12593	(3.21 $\times$)
timing terms together	2.00186	(5.70$\times$)

Table 3: Four curves that all first exceed 0.5 at frame 75 and never fall below it afterwards; the three step-like ones hold identical values at frames 74 and 75. Under the published definition they must score identically. They span 0.09093.

curve	private score	vs. step
graded run-up (0 $\rightarrow$ 0.49, then 0.51)	1.12964	+0.02519
step (0.49/0.51)	1.10445	reference
graded throughout (0.000 $\rightarrow$ 1.000)	1.06725	−0.03720
graded run-out (0.49, then 0.51 $\rightarrow$ 0.99)	1.03871	−0.06574
step at frame 76 (one-frame calibration)	1.08779	−0.01666

We quote the difference before the ratio deliberately. The metric is a weighted sum with no meaningful origin, so a ratio of two scores can be made to say almost anything by shifting the scale. The difference of 0.14874 and the decomposition below are scale-free; that the constant reaches 94.1% of the winning score is a convenience, not evidence.

One further row is worth stating plainly. The organisers’ own sample submission, resubmitted byte for byte, scores 1.04203, while the benchmark row displayed on the leaderboard scores 0.58333 — so those are not the same file, and the sample submission is not the origin of the published benchmark score.

4.2 Decomposing the score

Because every video-blind submission is identical across clips, all of them induce the same ordering over clips and earn the same AP and the same AUC, which therefore cancel in any difference between two of them (Section 3.4). Differencing the probes separates the terms; Table 2 gives both the probe scores and the components recovered from them.

Read the ratio precisely. It compares the timing terms at

their maximum against the discrimination terms at chance, not maximum against maximum. The maxima ratio is not identifiable from outside, because the weights are undisclosed, but it is bounded: the winner’s discrimination terms are worth at least $0.14874 + 0.35105 = 0.49979$, giving a maxima ratio of at most 4.01. Both framings say the same thing, and the measured statement is the safer one — a submission that reads nothing collects 2.00186, and the best discrimination anyone demonstrated on this benchmark added 0.14874 to it.

Two further readings follow. A constant at 0.51 and a constant at 0.99 score identically, so the metric reads only whether the curve stands above the threshold and never by how much. And the dip probe scores 1.85347 where the per-frame stable-timing coefficient predicts 1.79041, so the additive split of the timing total into its two parts is close but not exact; we report the split with that tolerance attached. The total itself is measured directly between two flat curves and does not depend on the split.

4.3 The score is linear in the crossing frame

Over the region in which the alarm still precedes the accident in every clip, the score falls by 0.016647 for each frame of delay — one point per 60.1 frames, or 0.4994 per second at 30 fps (Fig. 1). A dedicated one-frame experiment confirms the slope directly: a step at frame 75 scores 1.10445 and a step at frame 76 scores 1.08779, a difference of 0.01666 against $1/60 = 0.016667$.

The line was then asked to predict curves it had not seen. Three logistic curves crossing at frames 60, 75 and 90 are predicted to within 6.4×10^{-3} , the closest to 6×10^{-5} . A fourth, crossing at frame 105, lies beyond the fitted region where the curve has already begun to bend; the model is not expected to hold there and we report it rather than omit a held-out point that missed.

4.4 The published definition does not reproduce the scorer

One result does not fit, and we report it rather than smooth it. A linear ramp crosses the threshold at frame 75, exactly as a step function does, and scores 0.037 lower. That gap is 2.24 frames of slope, not an integer, so it cannot be a crossing-frame effect. We therefore submitted a diagnostic family holding the crossing frame at 75 and varying only the shape away from it (Table 3).

The four curves at the head of that table cross at the same frame and never fall back, and the three step-like ones hold identical values at frames 74 and 75. Under the metric as published they must score identically: both timing terms are functions of threshold crossings alone, and video-blind curves are identical across clips, so the discrimination terms cannot separate them either. They

span 0.09093, which is 5.5 frames of slope.

We do not have an explanation and state the two candidates rather than choose. Either the discrimination terms are not invariant to curve shape among video-blind submissions — which would weaken the cancellation argument of Section 3.4, though it is hard to reconcile with a step at frame 140 and a flat 0.49 scoring identically despite very different frame orderings — or the timing terms read something beyond the first crossing that the three agreeing curves share and the ramp does not.

Either way one conclusion is available without resolving it, and it is a finding in its own right: **the benchmark’s published formula does not reproduce its scorer.** An entrant cannot compute this competition’s score from its stated definition even given the labels. The headline result is unaffected, because the constant and the floor are both flat curves whose difference is measured directly; what carries a caveat is the split of the timing total.

4.5 A property of the withheld ground truth, recovered

For a step at frame k , both timing terms equal $\max(t_{ai} - k, 0)$ on each contributing clip, so the score above the floor is $c\mathbb{E}[\max(t_{ai} - k, 0)]$ with c the sum of the two timing weights. Differentiating, the slope at k is $-cP(t_{ai} > k)$: the slope measures the survival function of the accident-onset distribution, scaled. The leaderboard is therefore reporting, one submission at a time, the shape of an annotation no entrant has seen.

The initial segment estimates c at 0.016667, and dividing the timing total by c gives a mean onset of 120.1 frames — about four seconds into a five-second clip. This explains mechanically why the constant does so well here: because the accident is always late in the window, an alarm at frame 0 is credited with roughly 120 frames of anticipation on every clip.

Two caveats belong with it. Under the model the magnitude of the slope must be non-increasing in k , because a survival function cannot rise; it is not, by roughly ninety times what five-decimal rounding permits, and we suspect the same second-order effect as in Section 4.4. And when k exceeds t_{ai} the maximum in the published definition is taken over an empty set and is undefined; we assume the scorer takes it as zero. We deliberately do not separate the weights from the metric values: every quantity above is a product of the two, and the positive rate cancels identically out of the onset ratio, so this measurement says nothing about what fraction of the corpus contains an accident.

4.6 The case study: our own entry

We reached this question by entering the competition, with a training-free ensemble combining CLIP-based semantic scoring, optical-flow motion energy and a third channel, fused by grid-searched weights with monotone clamping

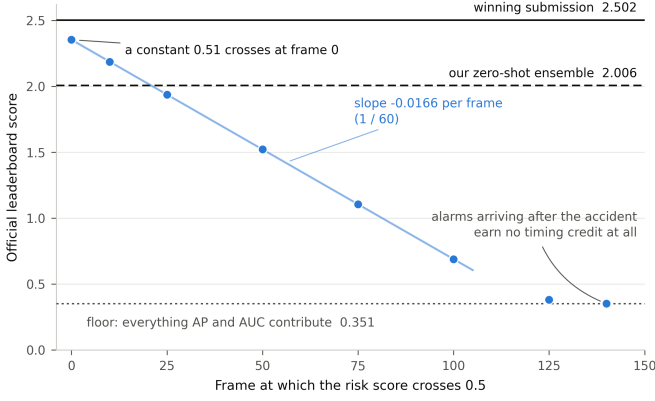


Figure 1: The official score is a straight line in the frame at which the alarm fires. Points are video-blind submissions scored by the organisers; the fit is over crossings up to frame 100. A constant crosses at frame 0.

and temporal post-processing. It placed ninth of thirteen at 2.00585 — *below the video-blind constant*. Auditing it against its released code rather than against our description of it found three defects of exactly the kind this paper is about. The third “semantic” channel is not a captioning model: it thresholds the same frame-difference statistic the motion channel uses and selects one of three fixed strings, so the ensemble has two independent inputs rather than three — which is why that channel posts an *earlier* mean crossover than the full ensemble. The monotone clamp forces precisely the condition STTA tests, so any report of STTA compliance measures that the clamp executed. And a temporal resampling stage evaluated the curve at index αt while emitting it at t , so the score reported for frame t was computed from a frame that had not yet arrived; a six-frame gain once attributed to it is withdrawn, and the released configuration disables the stage. No measurement above depends on this system. It is here because a critique of evaluation practice written by authors who had not audited their own entry would be worth less.

5 Validation on a labelled corpus

The competition publishes no labels. That is what makes its metric exploitable from outside, and it is also what makes the corrected protocol of Section 7 untestable there: every repair in it is stated in terms of quantities — a false-positive rate, an annotated collision time, a fixed operating point — that the benchmark does not expose. A protocol that cannot be run is a proposal, not a result.

This section runs it, on a corpus that is independent of the competition and fully annotated: the Nexar collision-prediction training split of 1,500 dashcam clips, 750 of which contain a collision and 750 of which are ordinary driving. Every positive clip carries two annotations, a col-

lision time t_{ev} and an alert time t_{al} marking the earliest moment at which a driver could still have acted. Three things follow that the competition does not allow. Average precision and area under the curve become measured quantities rather than a floor recovered by differencing. Earliness is measured to the annotated collision rather than to the end of the window. And because half the corpus contains no collision at all, the cost a warning imposes on ordinary driving becomes a number rather than an argument.

5.1 Setup

The video-blind family of Section 3.3 is regenerated for clips that differ in length and frame rate: each curve is a closed-form expression in the frame index, and a generator is given only the clip’s frame count and frame rate, so a curve that wanted a label could not read one. The published metric is implemented exactly as Section 3.2 states it, with t_{ai} resolved to the annotated collision in each clip’s own timebase.

Against these we place a sighted baseline. It is deliberately modest and no claim here depends on its accuracy; its only job is to read pixels, so that the protocol has something sighted to separate from the blind. Five motion statistics are computed per frame at a fixed 7.5 Hz — bulk and localised frame difference, frame difference restricted to the lower-centre of the image, luminance and its change — and a logistic model over causal summaries of them predicts whether the current moment lies inside the annotated actionable window $[t_{al}, t_{ev}]$. The split is 600 clips for training and 300 held out, fixed before any modelling. Every feature at time t is a function of samples at or before t ; the property is asserted by perturbation rather than assumed, by requiring that altering the tail of a clip leave every earlier feature bit-identical.

Two properties of the corpus are worth stating before any score, because both bear on the argument. Its clips carry **36 distinct frame rates** between 23.6 fps and 31.0 fps, so an earliness term counted in frames is summing quantities of different duration within a single dataset. And its collisions occur between 3.03s and 56.80s into their clips, an **18.7-fold spread** in the largest earliness any submission can collect. The scale objection of Section 6(a) does not require two differently windowed corpora to demonstrate; one is enough.

5.2 The metric behaves the same way, and the corpus leaks

Table 4 scores the video-blind family on all 1,500 clips. A constant risk score of 0.51 collects 19.09s of TTA@0.5 and 19.09s of STTA@0.5 — **99.9% of the 19.10s ceiling** the corpus admits — while scoring exactly 0.500 on both discrimination terms, which is what a constant must score. The result of Section 4 is therefore not a property of one

organiser’s scorer. It follows from the composition, and it reappears wherever the composition is used.

One finding here was not anticipated by the analysis of Section 6 and could not have been made on the competition corpus. Clip duration alone separates collisions from ordinary driving at an AUC of 0.537; frame count alone at 0.548; and **the frame rate of the video file alone at 0.533**. A submission that opens the container header, reads how fast the file was encoded, and never decodes a frame scores above chance.

This is a property of how the corpus was assembled rather than of the metric, but it has a direct consequence for anyone evaluating on it: a discrimination score slightly above chance is not evidence that a method has seen anything. We therefore add a *metadata-only* control — a model given the file’s frame rate and duration and nothing else — to the protocol of Section 7, alongside the video-blind one. We did not know to ask for it before running this study, which is itself the argument for running controls rather than reasoning about them.

5.3 What each submission earns on held-out clips

Table 5 scores everything on the same 300 held-out clips. Under the published metric the ordering is inverted with respect to what the metric is supposed to reward. The sighted model is the only submission that discriminates at all, at AUC 0.671 against 0.500 for every blind curve, and it earns 0.045 s of TTA. The constant, at chance discrimination, earns 18.86 s. **The composition pays blindness 424 times better than sight.**

The mechanism is the one Section 6(c) identifies: neither timing term reads the magnitude of a score, only whether it stands above 0.5. A model trained to be calibrated on a corpus where roughly one frame in fifty is inside an actionable window produces probabilities that rarely cross a threshold set at one half, and is penalised for it. A submission that is simply pinned above the threshold is not.

5.4 The protocol, tested and found insufficient

Section 7 makes a specific and falsifiable prediction about its own two halves: bounding the earliness term normalises it into the unit interval but *does not* remove the constant as maximiser, and only conditioning on a fixed operating point of the discrimination metric does.

That prediction holds exactly. Under bounding alone the constant scores 1.0000 — it still takes the whole of the available anticipation, now expressed as a ratio. Conditioned at a false-positive rate of 10% it scores 0.0000, because there is no threshold at which it detects anything without detecting everything. Had the constant failed at

the bounding step instead, either the implementation or the analysis of Section 6 would have been wrong.

The protocol does not survive the metadata-only control, and it fails under both readings of its own averaging rule. Read as written — normalised earliness averaged over the positives a submission detects — the metadata-only control scores a **perfect** 1.0000 against the sighted model’s 0.0611, while detecting 14 of 150 positives to the sighted model’s 35. Averaged instead over all positives, charging an undetected one zero, it scores 0.0933 against 0.0143. Either way the submission that reads the file header outscores the one that reads the road, under the protocol proposed to prevent exactly this.

The reason is structural and we had not seen it. Conditioning uses a *clip-level* discrimination metric, which asks whether a submission ranks collision clips above ordinary ones. It cannot ask whether the alarm, within a clip, is raised at a useful moment. A curve that is flat across a clip passes clip-level conditioning on whatever discrimination it has, and then collects maximal earliness on every clip it detects at all, because a flat curve above threshold has been above threshold since the first frame.

The two readings above are the second defect. Averaging over detected positives only — the natural reading of Section 7, item 2 — lets a submission buy a perfect score by detecting almost nothing, which is precisely what the metadata-only control did. Earliness that is never delivered, because the alarm was never raised, is not earliness; the statistic must be averaged over all positives. That repair is necessary and, as the figures above show, nowhere near sufficient.

5.5 The repair: measure the alarm, not the ranking

Both defects have the same root. The protocol conditions on a quantity computed per clip, while the thing a warning system does — and the burden it imposes — happens per frame. The repair is to measure the alarm itself. Partition every frame of the corpus into two populations:

- **actionable:** frames of a collision clip inside $[t_{al}, t_{ev}]$, the interval the annotators mark as still avoidable;
- **non-actionable:** every frame of every ordinary-driving clip, plus frames of a collision clip before t_{al} .

Frames after the collision belong to neither: nothing is being anticipated there, and charging them would reward a curve for switching off after an event it cannot know has occurred. Then report two rates at a common threshold — *coverage*, the share of actionable time during which the alarm is raised, and *duty cycle*, the share of non-actionable time during which it is raised — and summarise by coverage at a capped duty cycle.

Fig. 2 sweeps both over every threshold. The diagonal is the no-skill line, and the permanent alarm sits on it, at

Table 4: The video-blind family scored on all 1,500 labelled Nexar clips, with earliness measured to the *annotated* collision. A constant above threshold takes 99.9% of the 19.10s the corpus admits while scoring exactly chance on both discrimination terms.

submission	AP	AUC	TTA@0.5 (s)	STTA@0. (s)
constant 0.51	0.500	0.500	19.09	19.09
constant 0.99	0.500	0.500	19.09	19.09
linear ramp 0→1	0.500	0.500	0.38	0.38
step at mid-clip	0.500	0.500	0.38	0.38
crosses, dips, recovers	0.500	0.500	19.09	13.10
crosses, then drops	0.500	0.500	19.09	0.01
never crosses (0.49)	0.500	0.500	0.00	0.00

the top right corner: it covers every actionable moment and alarms through every other one. That corner is the point the published composite scores at its maximum.

Table 5 gives the summary. At a duty cycle capped at 10% the entire video-blind family scores 0.0000, the metadata-only control 0.0855, and the sighted model 0.2833. The ordering is finally the one the metric was meant to produce, and it is produced by asking a question the published composite never asks.

The same partition answers in plain units the question Section 6 could only pose. A constant above threshold alarms for **60 minutes in every hour** of ordinary driving. The linear ramp alarms for 20.5 minutes per hour, the metadata-only control for 47.7, and the sighted model for 0.13s. None of this is visible to the published composite, whose timing terms are defined only on clips where a collision occurs, so the burden falls entirely on the discrimination half — measured here at 0.500 for every submission that pays it.

6 Why the Composition Fails

The scope of what follows is fixed before the argument rather than after it. This is an analysis of a metric’s algebra, illustrated by measurements on two corpora. Properties (a) and (b) are consequences of the definitions quoted in Section 3.2 and hold wherever those definitions are used. Property (c) is a statement about what the algebra implies, not a survey finding.

The defect is not a badly chosen weight. It is an error of type. Write the score as $B + U$, where B is the weighted sum of AP and AUC and is bounded above by the sum of their weights, and U is the weighted sum of the timing terms, bounded above by the sum of their weights times the mean accident onset — a quantity that depends on how the corpus was windowed.

(a) *The unbounded term grows with clip length while the bounded one does not.* The ratio of their maxima scales lin-

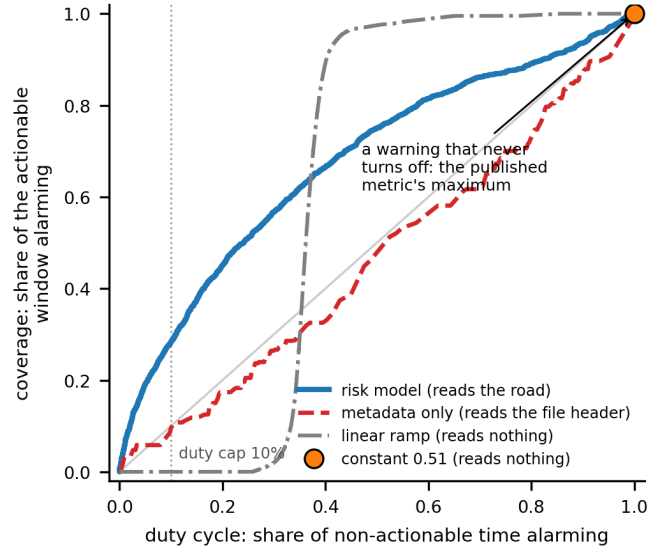


Figure 2: The warning operating characteristic: coverage of the annotated actionable window against duty cycle over all non-actionable time, swept over every threshold. The diagonal is the no-skill line. A permanent alarm sits on it, in the top right corner — the point the published composite scores at its maximum.

early in the mean onset, so two groups adopting the same weights on corpora windowed differently are not using the same metric, though they report the same name. On the competition the maximum of U is 2.00186, measured; the maximum of B is at least 0.49979, giving a ratio of at most 4.01, and against B at chance the measured ratio is 5.70. Section 5 shows the effect does not even need two corpora: within one, collisions ranging from 3.03 to 56.80s give an 18.7-fold spread in the earliness a single constant can collect.

(b) *The unbounded term has a constant as its global maximiser.* Any *unconditioned* threshold-crossing earliness measure is maximised by crossing immediately and never returning. The maximiser reads no input, so the term it maximises can carry no information about the input. The qualifier is the seed of the remedy: an earliness measure that is conditioned — evaluated at a fixed operating point of the discrimination metric, or penalised by the false-alarm rate — is not maximised by a constant.

(c) *Consequently the separating power of the score lies in B .* To the extent that a submission crosses early and holds, its U is at or near maximum and it is separated from other such submissions only by B . We cannot verify this for any particular entry, having no team’s curves but our own, so we state it as what the algebra implies and not as an observation about anyone’s system.

Property (c) is why the failure is hard to notice from inside. A leaderboard of this shape still orders methods usefully, because B still varies. What is invisible without

Table 5: All submissions on the same 300 held-out clips. Under the published metric (columns 2–3) the only submission that discriminates earns 0.04s of earliness and the one that reads nothing earns 18.86s. Bounding leaves the constant at its maximum, exactly as Section 7 predicts; conditioning removes it but is beaten by the metadata-only control. Only the frame-level statistic (last column) orders the submissions by what they can actually see.

submission	published metric		corrected protocol		frame-level alarm	
	AUC	TTA@0.5 (s)	bounded @0.5	conditioned @FPR 10%	alarm (min/h)	coverage @duty 10%
constant 0.51	0.500	18.86	1.000	0.000	60.0	0.000
linear ramp $0 \rightarrow 1$	0.500	0.30	0.019	0.000	20.5	0.000
step at mid-clip	0.500	0.31	0.019	0.000	20.5	0.000
never crosses (0.49)	0.500	0.00	0.000	0.000	0.0	0.000
metadata only	0.504	14.56	0.733	0.093	47.7	0.085
risk model (reads video)	0.671	0.04	0.003	0.014	0.0	0.283

Given the video file’s frame rate and duration. It never decodes a frame.

a control is that the number attached to each of them is dominated by a term available for free, so the reported score — the thing that goes into a table and gets compared across publications — is mostly a constant.

Two mechanisms compound this, and both are visible in our own pipeline rather than anyone else’s. *Metric-enforcing post-processing*: an operation that clamps the risk score above the threshold, followed by a report of stable-anticipation compliance, has measured only that the clamp executed. Stated generally, a post-hoc operation that enforces a metric’s definition renders that metric vacuous. *Non-causal timing gain*: an operation that resamples the curve so that frame t reports a value computed from a later frame buys time-to-accident from the future, and whatever earliness it buys cannot be obtained by any system running in real time. A third and milder mechanism is selection — reporting the mean anticipation of the best fifty clips of 1,417 is a statement about the selection, not the method.

6.1 What this costs safety research

Read as a system rather than as a curve, the maximiser of (b) is a device that alerts continuously, on every drive, whether or not anything is in front of the vehicle. Alarm timing is among the things that determine whether a driver trusts a forward collision warning and acts on it: warnings arriving when the driver does not expect them reduce that trust substantially [3], and a warning that is always present is the limiting case. A composite whose earliness half is maximised by permanent alarm is therefore not merely uninformative about warning capability; on this dimension it is ordered the wrong way round. Section 5 puts a number on the cost the metric cannot see: 60 min of alarm in every hour of ordinary driving.

A published anticipation figure therefore does not, on its own, establish that a system anticipates. A reader given only a composite score cannot tell how much was earned by

the method and how much was available to any submission that crosses early — and neither can the authors, unless a control was run.

7 A Corrected Protocol

Each item is traced to a mechanism above. The first is the one we would keep if we could keep only one. Items 1–6 were proposed before the validation of Section 5; items 7 and 8 exist *because* of it, and are the substantive change this paper makes to its own recommendation.

1. **Report a video-blind control with every anticipation result.** Compute the full metric on a constant above threshold, a linear ramp and a mid-clip step, and put those rows in the results table. If the control lands within noise of the method, the metric is not measuring the method. This costs one evaluation run, and should be as reflexive as a majority-class baseline in classification.
2. **Bound the earliness term, and condition it.** Bounding — normalising per clip so achieved over achievable lies in $[0, 1]$ — stops the term dominating the sum and stops it depending on the windowing. It does *not* remove the constant as maximiser, which still scores 1 and still gets it free; Section 5 confirms this directly. Conditioning on a fixed operating point is what removes the exploit. If only one is adopted, adopt conditioning.
3. **State the reference point of every timing metric:** the annotated collision, or the end of the clip. Reporting the interval to the end of the window under the name time-to-accident inflates every result by a corpus-dependent constant.
4. **Report discrimination alongside any timing metric,** never timing alone, and state the ratio of

positive to negative clips. Without negatives there is no false-positive rate, and any rising curve achieves perfect recall.

5. **Report metrics with post-processing disabled as well as enabled.** If a number moves when a clamp is removed, the clamp was part of the measurement. If a number moves when a resampling stage is removed, check whether that stage reads a frame that has not yet arrived.
6. **Publish the score of a released control alongside the leaderboard.** Organisers can do in an afternoon what took us twenty-five submissions, and an entrant would then see at a glance how much of their score they earned. For competitions that withhold labels, publish the weights and the scorer too: withholding labels prevents overfitting, but withholding the weights does not prevent the metric being exploited — this paper exploited it comprehensively without them — while it does prevent the sanity check that would have surfaced the problem before the competition opened.
7. **Average earliness over all positives, not over detected ones.** Averaging only where the alarm fired lets a submission buy a perfect score by detecting almost nothing; in Section 5 a control scored a perfect 1.0000 on 14 positives of 150. Earliness never delivered is not earliness.
8. **Measure the alarm at frame level, not clip level.** Clip-level conditioning cannot see that a curve is flat *within* a clip, and in Section 5 a control reading only the video file’s header beat a model reading the road under items 1–6 alone. Report *coverage* — the share of the annotated actionable window during which the alarm is raised — against *duty cycle* — the share of all non-actionable time during which it is raised — and summarise as coverage at a capped duty. Report a **metadata-only control** beside the video-blind one, because corpus metadata can itself carry the label.

8 Limitations

The leaderboard measurements rest on one competition and one corpus, and the scorer is closed, so Section 4.4 records a discrepancy we could not resolve. The validation corpus is a second, independent dataset, but it is still one dataset, and its label leakage through file metadata (Section 5) is a property of that corpus whose prevalence elsewhere we have not measured. The sighted baseline is deliberately weak; it establishes that the repaired statistic separates sighted from blind, not where a strong method would fall on it. The actionable window we score against is

the annotators’ judgement, and a different annotation convention would move the coverage numbers though not the ordering. Finally, our claim about the absence of video-blind controls in the literature covers the works whose results tables we inspected directly, and is not a systematic audit.

9 Conclusion

A composite that adds a bounded discrimination term to an unbounded earliness term does not measure warning capability. On a live benchmark a constant risk score reading no pixels matched the eighth of thirteen teams, and the timing terms were worth 5.70 times what discrimination contributed to it. On an independent annotated corpus the same constant took 99.9% of the available earliness at exactly chance discrimination, while the only submission that discriminated earned 0.045 of 18.86 seconds.

Three things are consequently cheap that were not. A video-blind control costs one evaluation run and settles whether a reported score is measuring a method or a metric. Probe submissions turn a leaderboard into a means of auditing a closed scorer, without labels and without the weights. And an earliness term that is bounded, conditioned on a fixed operating point, and measured at frame level against the time the alarm spends raised gives the field a quantity that can be compared across corpora windowed differently.

We would stress what the validation changed. The protocol we proposed was tested where ground truth exists, behaved exactly as predicted on the control it was designed against, and was then beaten by a control that never decodes a frame. A protocol that has not been run against adversarial controls is a hypothesis. What is at stake in getting this right is not tidiness: a warning system earns its place in a vehicle by alerting a driver early enough to act and seldom enough to be believed, and a metric whose earliness half is maximised by a permanent alarm cannot distinguish a system that does that from one that does nothing. Until anticipation results are reported beside a control that reads no video, a published warning horizon should be read as an upper bound on what the method might contribute, not as evidence of what it does.

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